

Context of the business

2Market is trying to understand the demographic of its customer, their behaviour and which marketing channel seems to sell the best and if there is any correlation with demographics. The business problem relies optimising marketing to generate an increase in sales and on ROI (return on investment) from marketing expenditure.

For further analysis and to better understand the project I would like to know:

- what is the board of the company going to do with the demographic and customer analysis we are going to provide them?
- Is there any additional data regarding which items are purchased through which channels?
- What is 2Market selling that directly targets children?

Analytical approach

I first clean data in SQL and Excel, I imported the raw data into Excel and examine the consistency. I formatted the columns in number, date and added a column to calculate the age with a simple formula ($=2025 - \text{year_birth}$). I find no duplicates, which I search through the ID column, there were neither empty cell values.

I later looked into the marital status and find out some ambiguous data as YOLO, alone and absurd. I wasn't sure how to treat both YOLO and absurd and so I decided to get rid of both of them (they were 0.09% each of the total population). I thought that 'alone' was a problem of inconsistency and so converted those four as single.

I analysed the income and age column and removed outliers in both of them. Values exceeding 112k in income and customer of over 84 years old. Some of them were reaching 124 and over, so those might be some forgotten data.

I sort the data by age, sql, age related to the income, marital status and I found some relevant information. I didn't find some relevant correlation between age and income, neither between income and amount of items bought, but I found some interesting data distribution by country, education, marital status, income and country.

I've decided to cross-reference the data, starting with key characteristics as country, marital status, age and education. I chose to be more specific about marital status because there was some data regarding children which I could

leverage. I opted for marital status to extract as much information as possible from the data I had available and to connect to different variable.

I used the same clean data from Excel to SQL. I imported the data and I created two new tables through the column id and looked into the successful lead conversion of each social network, brochure and bulk e-mail, and the total successful of lead conversion. I further analyse the data grouping them by marital status, country, and education. The data from the two tables created by SQL gave me some clear information regarding the most effective advertising channels.

Dashboard design and development

I chose the dashboard to enlighten the data I had selected from the outcome of my analysis. The upper part of the dashboard answers the demographic main issue: revenues for each marital status, marital status percentage through the customer, a lean data about average income and family; both who have children and family who don't. These charts are followed the percentage of customer by education level. Last chart show us what is the successful conversion through different marketing medium.

The dashboard contains a lot of data, and a lot of number as well, so I decided to introduce just one interactive tool, which can highlight the percentage of the marital status and the amount that the selected marital status spends of the six categories of the most sold income. I chose a palette of blue for the martial status chart and revenue chart which added consistency and created a clear link to one and other. For the education chart I opted for a palette of red, and I selected different shades of black for the marketing chart. I decided to choose not to many colour to avoid any confusion and for a clear view.

I opted a simple and concise title to the charts which can directly give an idea of the content of the charts. A few colours were selected thinking about accessibility and I chose simple chart due to the large amount of data presented in the dashboard.

Patterns, trends, and insights

There are some clear insight coming out from this data analysis. The customers are mostly in their 40s, 50s and 60s. Half of the customers are married or they are in a relationship. They are well educated, with half of the customer who have obtained a graduate degree, and a notable segment of around 20% have a PhD, 15% have a master's. This is an interesting point; one may assume that because they are getting older they tend to have higher wages and thus they become more conscious consumer who can

focus more on the quality of the product. Or at least we should investigate the reason behind the high percentage of well-educated customer. Half of the customers are from Spain while a consistent percentage of 15% and 10% are from South Africa and Canada. They are followed by Australia, Germany, India, and United States, each with around 5%. So it would be interesting to know if in 2Market shop are available items from the country of origin. Half of the product bought by our customer are liquor, followed by 27% from meat, and then small by commodities, fish, chocolate and vegetables. A key characteristic is the fact that around 50% of the customer have at least a kid or a child but there are no relevant kids-oriented product sold to the customer, except for chocolate.

Regarding the marketing issue there is far better effectiveness in reaching people through social network and bulk e-mail instead of brochure. The total number of lead conversion on the internet, whether is a social network or bulk e-mail, is five times compared to brochures. If additional data were available, it would be interesting to investigate the total cost per each conversion (pay-per-click for each medium) compared to the cost of a customer purchase through brochure, and would be interesting to investigate what kind of items was bought through which canal.